



CONTINUOUS PROBABILITY DISTRIBUTION

PART A: CONTINUOUS PROBABILITY

2 unit

selective

Q 1

Ace

The probability density function for the continuous random variable X is given by:

$$f(x) = \begin{cases} |1-x| & 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

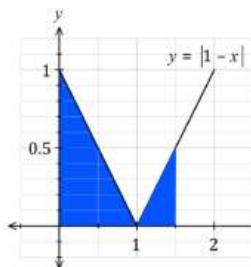
(a) Find $P(X \leq 1.5)$

(b) Find $P(1 \leq X \leq 2)$

(a) $P(X \leq 1.5)$

$$= \frac{1}{2} \times 1 \times 1 + \frac{1}{2} \times 0.5 \times 0.5$$

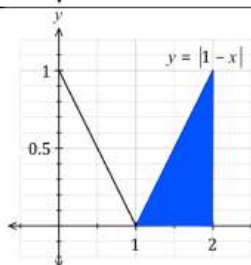
$$= 0.625$$



(b) $P(1 \leq X \leq 2)$

$$= \frac{1}{2} \times 1 \times 1$$

$$= 0.5$$



Q 2

Ace

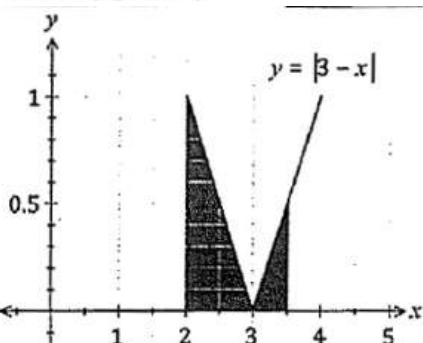
A continuous random variable X has a function f given by

$$f(x) = \begin{cases} |3-x| & 2 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find $P(2 \leq X \leq 3.5)$

(b) Find $P(2 \leq X \leq 2.5)$

(a)



$$P(2 \leq X \leq 3.5) = \frac{1}{2} \times 1 \times 1 + \frac{1}{2} \times 0.5 \times 0.5$$

$$= 0.625$$

(b)

$$P(2 \leq X \leq 2.5) = \frac{1}{2} \times 0.5 \times (1 + 0.5)$$

$$= 0.375$$

Q 3

Baulkham Hills

Given the probability density function

$$f(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find the cumulative distribution function $F(x)$.

(ii) Hence find the median.

$$F(x) = \int_0^x 2e^{-2x} dx$$

$$F(x) = 2x - \frac{1}{2} [e^{-2x}]_0^x$$

$$F(x) = -[e^{-2x} - e^0]$$

$$\therefore F(x) = -e^{-2x} + 1$$

(iii) let m be median

$$\therefore F(m) = \frac{1}{2}$$

$$\frac{1}{2} = -e^{-2m} + 1$$

$$e^{-2m} = 1 - \frac{1}{2}$$

$$\ln(e^{-2m}) = \ln \frac{1}{2}$$

$$-2m = \ln \frac{1}{2}$$

$$m = -\frac{1}{2} \ln \frac{1}{2} \text{ (median)}$$

$$\therefore \text{median} = -\frac{1}{2} \ln \frac{1}{2} \text{ or } \ln \sqrt{2}$$

$$\text{or } 0.347$$

$$\text{or } -\frac{1}{2} (-\ln 2) = \frac{1}{2} \ln 2$$

Q 4

Birrong Girls

A probability density function can be used to model the lifespan of a termite, X , in weeks, is given by

$$f(x) = \begin{cases} k(36 - x^2) & 3 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

(i) Show that the value of k is $\frac{1}{45}$ (2 marks)

(ii) Find the cumulative distribution function. (2 marks)

(iii) Find the probability that a termite's lifespan is greater than 5 weeks. (1 mark)

$$\int_3^6 k(36 - x^2) dx = 1$$

$$k \left[36x - \frac{x^3}{3} \right]_3^6 = 1$$

$$\left[36(6) - \frac{6^3}{3} \right] - \left[36(3) - \frac{3^3}{3} \right] = \frac{1}{k}$$

$$45 = \frac{1}{k} \quad k = \frac{1}{45}$$

$$F(x) = \int_3^x f(x) dx$$

$$= \int_3^x \left(\frac{36 - x^2}{45} \right) dx = \frac{1}{45} \left[36x - \frac{x^3}{3} \right]_3^x$$

$$= \frac{1}{45} \left[\left(36x - \frac{x^3}{3} \right) - \left(36(3) - \frac{3^3}{3} \right) \right]$$

$$= \frac{1}{45} \left[36x - \frac{x^3}{3} - 99 \right] = \frac{4x}{5} - \frac{x^3}{135} - \frac{11}{5}$$

$$P(X > 5) = 1 - P(X \leq 5)$$

$$= 1 - \left(\frac{4(5)}{5} - \frac{5^3}{135} - \frac{11}{5} \right) = \frac{17}{135}$$





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2 unit



Q 5

Blacktown Boys

The length of time, in minutes, that a customer queues to buy toilet paper is a random variable, X , with probability density function

$$f(x) = \begin{cases} k(64 - x^2) & \text{for } 0 \leq x \leq 8 \\ 0 & \text{otherwise} \end{cases}$$

where k is a constant.

- Show that the value of k is $\frac{3}{1024}$.
- Find the cumulative distribution function.
- Find the probability that a customer will queue for longer than 5 minutes, round your answer to 4 significant figures.

$$\begin{aligned} \text{ia } \int_0^8 k(64 - x^2) dx &= 1 \\ k \left[64x - \frac{x^3}{3} \right]_0^8 &= 1 \\ k \left[64 \times 8 - \frac{8^3}{3} - 0 \right] &= 1 \\ k \times \frac{1024}{3} &= 1 \\ k &= \frac{3}{1024} \end{aligned}$$

$$\begin{aligned} F(x) &= \frac{3}{1024} \left(64x - \frac{x^3}{3} \right) \\ F(x) &= \frac{3x}{16} - \frac{x^3}{1024} \\ \therefore F(x) &= \begin{cases} 0 & x < 0 \\ \frac{3x}{16} - \frac{x^3}{1024} & 0 \leq x \leq 8 \\ 1 & x > 8 \end{cases} \end{aligned}$$

$$\begin{aligned} \text{ib } f(x) &= \begin{cases} \frac{3}{1024}(64 - x^2) & \text{for } 0 \leq x \leq 8 \\ 0 & \text{otherwise} \end{cases} \\ F(x) &= \int_0^x \frac{3}{1024}(64 - x^2) dx \\ F(x) &= \frac{3}{1024} \left[64x - \frac{x^3}{3} \right]_0^x \\ F(x) &= \frac{3}{1024} \left[64x - \frac{x^3}{3} \right]_0^x \end{aligned}$$

$$\begin{aligned} \text{ic } P(X > 5) &= 1 - P(X \leq 5) \\ P(X > 5) &= 1 - F(5) \\ P(X > 5) &= 1 - \left(\frac{3}{16} \times 5 - \frac{5^3}{1024} \right) \\ P(X > 5) &= \frac{189}{1024} \\ P(X > 5) &= 0.184570... \\ P(X > 5) &= 0.1846 \quad (4 \text{ sig figs}) \end{aligned}$$

Q 6

East Hills Boys

A function is given by $f(x) = \begin{cases} \frac{3x^2}{125} & 0 \leq x \leq 5 \\ 0 & \text{elsewhere} \end{cases}$

If this function is a continuous probability distribution, what is the area under the curve?

- 1
 - 0.5
 - 1
 - 2

Q 7

Killara

The queueing time, X minutes, of a customer at the checkout of a supermarket has the probability density function

$$f(x) = \begin{cases} \frac{12}{625} x^2(k - x) & 0 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$

- Show that the value of k is 5. 3
- Is a customer more likely to wait less than 3 minutes, or more than 3 minutes? Justify your answer with calculations. 3

$$\begin{aligned} \text{(a) } \int_0^k \frac{12}{625} x^2(k - x) dx &= 1 \\ \frac{12}{625} \int_0^k kx^2 - x^3 dx &= 1 \\ \left[\frac{kx^3}{3} - \frac{x^4}{4} \right]_0^k &= \frac{625}{12} \end{aligned}$$

$$\begin{aligned} \frac{k^4}{3} - \frac{k^4}{4} &= \frac{625}{12} & \frac{k^4}{12} &= \frac{625}{12} \\ k^4 &= 625 & \therefore k &= \sqrt[4]{625}, k \geq 0 & k &= 5 \end{aligned}$$

$$\begin{aligned} \text{(b) } P(X \geq 3) &= \int_3^5 \frac{12}{625} x^2(5 - x) dx \\ &= \frac{12}{625} \int_3^5 5x^2 - x^3 dx \\ &= \frac{12}{625} \left[\frac{5x^3}{3} - \frac{x^4}{4} \right]_3^5 \\ &= \frac{12}{625} \left[\left(\frac{625}{3} - \frac{625}{4} \right) - \left(45 - \frac{81}{4} \right) \right] \\ &= \frac{12}{625} \left[\frac{625}{12} - \frac{99}{4} \right] = \frac{328}{625} = 0.5248 \end{aligned}$$

This means that a customer is more likely to wait more than 3 minutes.



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Q 8

North Sydney Girls

The continuous random variable X has cumulative distribution function F given by

$$F(x) = \begin{cases} 0 & \text{for } x < 1, \\ k(x^2 - x) & \text{for } 1 \leq x \leq 2, \\ 1 & \text{for } x > 2. \end{cases} \quad \text{where } k \text{ is a constant.}$$

- (a) Show that $k = \frac{1}{2}$, and hence find the 90th percentile value of X , giving your answer correct to three significant figures.
- (b) Find an expression for $f(x)$ which represents the probability density function of X .

(a) $F(2) = 1$
 $\therefore 1 = k(2^2 - 2)$
 $1 = k \times 2$
 $k = \frac{1}{2}$
 $0.9 = \frac{1}{2}(x^2 - x)$
 $1.8 = x^2 - x$
 $0 = x^2 - x - 1.8$
 $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-1.8)}}{2}$

$= \frac{1 \pm \sqrt{1 + 7.2}}{2} = 1.932, -0.932$
 Since $1 \leq x \leq 2$ $x = 1.932$ (3dp)
 (b) $f(x) = F'(x) = \frac{1}{2}(2x - 1)$
 $= x - \frac{1}{2}$
 $\therefore f(x) = \begin{cases} x - \frac{1}{2} & 1 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$

Q 9

Sydney Tech

For the following continuous probability distribution

$$f(x) = \begin{cases} \frac{3x^2}{511} & \text{for } 1 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find $P(X = 5)$
 (b) Find $P(X \leq 5)$
 (c) Find $P(X > 5)$

(a) $P(X = 5) = 0$
 (b) $P(X \leq 5) = \int_1^5 \frac{3x^2}{511} dx$
 $= \frac{1}{511} \int_1^5 3x^2 dx = \frac{1}{511} [x^3]_1^5$
 $= \frac{1}{511} [5^3 - 1^3] = \frac{1}{511} [124]$
 $= \frac{124}{511}$
 (c) $P(X > 5) = 1 - P(X \leq 5)$
 $= 1 - \frac{124}{511} = \frac{387}{511}$

Q10

Total Education

A continuous random variable, X , has a probability density function given by

$$f(x) = \begin{cases} \frac{k}{2x-1} & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Show that the value of k is $\frac{2}{\ln 7}$.
 (b) Find $P(X < 2)$.

(a) $\int_1^4 \frac{k}{2x-1} dx = 1 \Rightarrow \left[\frac{k}{2} \ln(2x-1) \right]_1^4 = 1 \Rightarrow \frac{k \ln 7}{2} = 1 \Rightarrow k = \frac{2}{\ln 7}$

(b) $\int_1^2 \frac{k}{2x-1} dx = \left[\frac{k}{2} \ln(2x-1) \right]_1^2 = \frac{k \ln 3}{2} = \frac{\ln 3}{\ln 7} = \log_7 3$ (since $k = \frac{2}{\ln 7}$)

Q11

Total Education

A function is defined as follows:

$$f(x) = \begin{cases} ax + b & x < 0 \\ e^{2x} & x \geq 0 \end{cases}$$

Find a and b if $f(x)$ is continuous and has a derivative at $x = 0$.

$$\begin{cases} f(0) = a \times 0 + b = e^{2 \times 0} \\ f'(0) = a = 2e^{2 \times 0} \end{cases} \Rightarrow a = 2, b = 1$$



Continuous Probability Distribution

Part b: 3m's and variance

2 unit

selective

Q 1

Ace

The students in a school were surveyed on the number of hours of sleep per week. The results were normally distributed. The survey indicated that 95% of students had between 42 and 54 hours of sleep per week.

- Determine the mean number of hours of sleep per week.
- What was the standard deviation?
- What percentage of students would have indicated they had between 51 and 57 hours of sleep per week?

(a) 95% of the data lie within two standard deviations of the mean.
42 hours is a z-score of -2 and 54 hours has a z-score of 2.
The mean is midway between 42 and 54 hours.
 $\therefore$ Mean is 48 hours.

(b) There are 4 standard deviations between 42 and 54 hours.
Standard deviation = $\frac{54 - 42}{4}$
= 3 hours

(c) To find the z-score of 51 and 57

$$z = \frac{x - \bar{x}}{s} = \frac{51 - 48}{3} = 1$$

$$z = \frac{x - \bar{x}}{s} = \frac{57 - 48}{3} = 3$$

$$\text{Percentage} = \frac{99.7\%}{2} - \frac{68\%}{2}$$

$$= 15.85\%$$

$\therefore$ 15.85% have sleep between 51 and 57 hours of sleep per week

Q 2

Ace

The probability density function for the continuous random variable X is:

$$f(x) = \begin{cases} x^3 - x + 4 & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{Find the } E(X).$$

To find the expected value or mean

$$\int_{-\infty}^{\infty} xf(x)dx = \int_0^1 x(x^3 - x + 4)dx$$

$$= \int_0^1 (x^4 - x^2 + 4x)dx = \left[\frac{x^5}{5} - \frac{x^3}{3} + 2x^2 \right]_0^1 = \frac{28}{15} = 1 \frac{13}{15}$$

Q 3

Baulkham Hills

Given a function $f(x) = \begin{cases} 6x - 6x^2 & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$

- Show that $f(x)$ represents probability density function. 2
- Find the mode of the probability density function $f(x)$. 1

(i) $f(x)$ represents PDF if

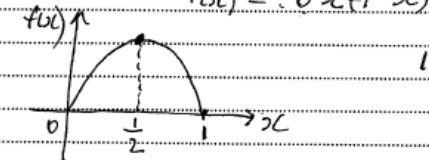
$$\bullet \int f(x) = 1 \text{ and } f(x) \geq 0$$

$$\therefore \int_0^1 (6x - 6x^2) dx$$

$$= \left[3x^2 - 2x^3 \right]_0^1$$

$$= [3 - 2] - 0 = 1 \therefore \text{Yes it's PDF}$$

(ii) By sketching $f(x) = 6x - 6x^2$
 $f(x) = 6x(1-x)$



Mode is $x = \frac{1}{2}$ with the highest value of $f(x)$

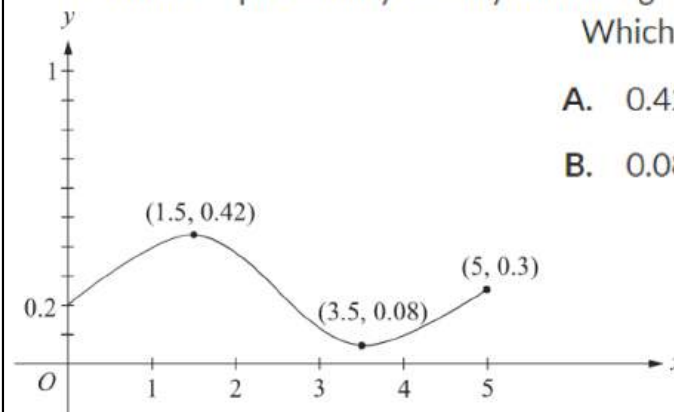
Q 4

Birrong Girls

A continuous probability density function graph is drawn below.

Which of the following is the mode?

- A. 0.42 C. 1.5
- B. 0.08 D. 5



2





Continuous Probability Distribution

Part b: 3m's and variance

2 unit

selective

Q 5

Birrong Girls

A continuous random variable, X , has a probability density function given by (3 marks)

$$f(x) = \begin{cases} \frac{1}{5} e^{-\frac{x}{5}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The median of X is m .
Determine the value of m .

$$\int_0^m \frac{1}{5} e^{-\frac{x}{5}} dx = \frac{1}{5} \left[-e^{-\frac{x}{5}} \right]_0^m$$

$$= \left[-e^{-\frac{x}{5}} \right]_0^m$$

$$= -e^{-\frac{m}{5}} + e^0$$

$$= 1 - e^{-\frac{m}{5}}$$

$$\therefore 1 - e^{-\frac{m}{5}} = 0.5$$

$$e^{-\frac{m}{5}} = 0.5$$

$$-\frac{m}{5} = \ln 0.5$$

$$m = -5 \ln(0.5)$$

$$\approx 3.47$$

$\therefore$ the median is 3.47

Q 6

Caringbah

The continuous random variable X has probability density function $f(x) = \frac{1}{2} \sin x$ for $0 \leq x \leq \pi$.

(a) Find the cumulative distributive function (CDF)

(b) Find the first quartile of the distribution.

$$(a) \int_0^x \frac{1}{2} \sin t dt = -\frac{1}{2} [\cos t]_0^x$$

$$= -\frac{1}{2} (\cos x - 1)$$

$$\therefore F(x) = -\frac{1}{2} (\cos x - 1)$$

$$(b) \frac{1}{2} (1 - \cos x) = \frac{1}{4}$$

$$1 - \cos x = \frac{1}{2}$$

$$\cos x = \frac{1}{2}$$

$$x = \frac{\pi}{3}$$

Q 7

Caringbah

The time t minutes that a customer spends waiting in a queue has a probability density function:

$$f(x) = \begin{cases} \frac{x(k-x)}{36} & \text{for } 0 \leq x \leq k \\ 0 & \text{otherwise} \end{cases}$$

(a) Show that the value of k is 6.

(b) Write down the value of $E(X)$.

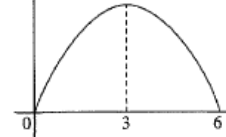
(c) Calculate $\text{Var}(X)$.

$$(a) \frac{1}{36} \int_0^k (kx - x^2) dx = \frac{1}{36} \left[\frac{kx^2}{2} - \frac{x^3}{3} \right]_0^k$$

$$= \frac{1}{36} \left[\frac{k^3}{2} - \frac{k^3}{3} \right] = \frac{1}{36} \left[\frac{3k^3 - 2k^3}{6} \right] = \frac{k^3}{216}$$

$$\text{so } \frac{k^3}{216} = 1 \quad k = \sqrt[3]{216} \quad k = 6$$

(b) $E(X) = 3$



$$(c) \text{var}(X) = E(X^2) - E(X)^2$$

$$E(X^2) = \frac{1}{36} \int_0^6 x^3 (6-x) dx = \frac{1}{36} \int_0^6 (6x^3 - x^4) dx$$

$$= \frac{1}{36} \left[\frac{6x^4}{4} - \frac{x^5}{5} \right]_0^6 = \frac{1}{36} \left[1944 - \frac{7776}{5} \right] = 10.8$$

$$\text{var}(X) = 10.8 - 9 = 1.8$$

Q 8

East Hills Boys

2 Find the median of the continuous probability distribution defined as $f(x) = \frac{x^3}{64}$ in the domain $[0, 4]$.

$$f(x) = \frac{x^3}{64}$$

Find CDF

$$\int_0^x \frac{x^3}{64} dx$$

$$= \frac{1}{64} \int_0^x x^3 dx = \frac{1}{64} \left[\frac{x^4}{4} \right]_0^x$$

$$= \frac{1}{64} \left[\frac{x^4}{4} - \frac{0^4}{4} \right] = \frac{1}{64} \left(\frac{x^4}{4} \right) = \frac{x^4}{256}$$

For Median

$$\int_0^x f(x) dx = 0.5$$

$$\frac{x^4}{256} = 0.5$$

$$x^4 = 128$$

$$x = 3.363585661$$

$$x = 3.4$$

$$P(X < 3.4) = 0.5$$



Continuous Probability Distribution

Part b: 3m's and variance

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Q 9

Gosford

The continuous random variable X has probability density function $f(x)$ given by

$$f(x) = \begin{cases} k(x^2 - 2x + 2) & 1 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases} \quad \text{Where } k \text{ is a constant}$$

- (a) Show that $k = \frac{1}{12}$
 (b) Fully define the cumulative distribution function $F(x)$.
 (c) Show that the median of X lies between $x = 3.2$ and $x = 3.3$

(a) $F(x) = k \int_1^x (x^2 - 2x + 2) dx$
 $\therefore k \left[\frac{x^3}{3} - x^2 + 2x \right]_1^x = 1$
 $\therefore k \left(\left(\frac{64}{3} - 16 + 8 \right) - \left(\frac{1}{3} - 1 + 2 \right) \right) = 1$
 $\therefore 12k = 1, k = \frac{1}{12}$

(b) For $1 \leq x \leq 4$
 $F(x) = \frac{1}{12} \int_1^x (x^2 - 2x + 2) dx$
 $= \frac{1}{12} \left[\frac{x^3}{3} - x^2 + 2x \right]_1^x$
 $= \frac{1}{12} \left[\frac{x^3}{3} - x^2 + 2x - \left(\frac{1}{3} - 1 + 2 \right) \right]$

(c) $F(3.2) = \frac{3.2^3}{36} - \frac{3.2^2}{12} + \frac{3.2}{6} - \frac{1}{9}$
 $= 0.479 < 0.5$
 $F(3.3) = \frac{3.3^3}{36} - \frac{3.3^2}{12} + \frac{3.3}{6} - \frac{1}{9}$
 $= 0.5296 > 0.5$
 $F(\text{median}) = 0.5$
 $\therefore \text{median lies between } x = 3.2 \text{ and } x = 3.3$

Q10

Hunters Hill

A continuous random variable X has a function f given by

$$f(x) = \begin{cases} 2x^3 - x + a, & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find the value of a which makes $f(x)$ a valid probability density function
 (b) Find the expected value and variance of X .

a) $F(x) = \int_0^1 (2x^3 - x + a) dx = 1$ if pdf.
 Since $F(x) = 0$ for all other x values
 $\left[\frac{x^4}{2} - \frac{1}{2}x^2 + ax \right]_0^1 = 1$
 $\frac{1}{2} - \frac{1}{2} + a = 1 \quad a = 1$

b) $E(X) = \mu = \int_0^1 x f(x) dx$
 $= \int_0^1 x(2x^3 - x + 1) dx$

$= \int_0^1 (2x^4 - x^2 + x) dx$
 $= \left[\frac{2}{5}x^5 - \frac{1}{3}x^3 + \frac{1}{2}x^2 \right]_0^1 = \frac{17}{30}$

$\text{Var}(X) = \sigma^2 = \int_0^1 x^2 f(x) dx - \mu^2$
 $= \int_0^1 (2x^5 - x^3 + x^2) dx - \left(\frac{17}{30} \right)^2$
 $= \left[\frac{1}{3}x^6 - \frac{1}{4}x^4 + \frac{1}{3}x^3 \right]_0^1 - \left(\frac{17}{30} \right)^2 = \frac{47}{450}$

Q11

James Ruse

Consider the probability density function f for a random variable X given by

$$f(x) = \begin{cases} \frac{a}{\sqrt{9+2x}} & 0 \leq x \leq 8 \\ 0 & \text{otherwise} \end{cases} \quad \text{where } a \text{ is a constant.}$$

- a) Find the value of a .
 b) Find the cumulative distribution function, $F(x)$.
 c) Hence find the interquartile range, $Q_3 - Q_1$.

a) Require $1 = \int_{-\infty}^{\infty} f(x) dx = \int_0^8 \frac{a}{\sqrt{9+2x}} dx$
 so $1 = a \int_0^8 (9+2x)^{-1/2} dx$
 $1 = a \left[(9+2x)^{1/2} \right]_0^8$
 $1 = a [5 - 3] \quad 1 = 2a \quad a = \frac{1}{2}$

b) $F(x) = \frac{1}{2} \int_0^x (9+2x)^{-1/2} dx$
 $= \frac{1}{2} \left[(9+2x)^{1/2} \right]_0^x$
 $= \frac{1}{2} (\sqrt{9+2x} - 3)$
 $F(x) = \begin{cases} 0 & \text{for } x < 0 \\ \frac{\sqrt{9+2x} - 3}{2} & \text{for } 0 \leq x \leq 8 \\ 1 & \text{for } x > 8 \end{cases}$

c) IQR = $Q_3 - Q_1$
 $F(x) = \frac{1}{2} (\sqrt{9+2x} - 3) \Rightarrow$
 $x = \left(2 \left(\frac{3}{2} + F(x) \right) \right)^2 - 9$
 Q_3 when $F(x) = \frac{3}{4}$
 $Q_3 = \left(2 \left(\frac{3}{2} + \frac{3}{4} \right) \right)^2 - 9 = \frac{45}{8} = 5.625$
 Q_1 when $F(x) = \frac{1}{4}$
 $Q_1 = \left(2 \left(\frac{3}{2} + \frac{1}{4} \right) \right)^2 - 9 = \frac{13}{8} = 1.625$
 IQR = $Q_3 - Q_1 = 4$



Continuous Probability Distribution

Part b: 3m's and variance

2 unit

selective

Q12

Manly

The monthly sales, X (in units of \$100, 000), of a business are modelled by the two probability density functions:

$$g(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{Model 1}$$

OR

$$f(x) = \begin{cases} kx^3(1-x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{Model 2}$$

- Determine the value of k in the second model.
- Show that the two models have the same median value.

$$\text{i} \quad \int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow k \int_0^1 (x^3 - x^5) dx = 1$$

$$\text{Since } \int_0^1 (x^3 - x^5) dx = \left[\frac{x^4}{4} - \frac{x^6}{6} \right]_0^1 = \frac{1}{4} - \frac{1}{6} = \frac{1}{12} \quad \text{then } \frac{k}{12} = 1 \Rightarrow k = 12$$

- Let $x = M$ be the median of model g . Then:

$$\frac{1}{2} = \int_0^M g(x) dx = \int_0^M 2x dx = \left[x^2 \right]_0^M = M^2, \text{ i.e. } M^2 = \frac{1}{2}$$

$$\text{Consider } \int_0^M f(x) dx = \int_0^M 12(x^3 - x^5) dx = \left[3x^4 - 2x^6 \right]_0^M = 3M^4 - 2M^6$$

$$\text{Since } M^2 = \frac{1}{2}, \int_0^M f(x) dx = 3\left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{2}\right)^3 = \frac{1}{2}$$

Hence $x = M$ is also the median of model f , i.e. the two models have the same median.

Q13

Manly

Let X be a continuous random variable whose probability density function is $f(x) = \frac{3}{4}(2x - x^2)$ for $0 \leq x \leq 2$. What is the mode of X ?

- A. 0 B. 0.5 C. 0.75 **D. 1**

$$f'(x) = 0 \Rightarrow \frac{3}{4}(2 - 2x) = 0 \Rightarrow x = 1$$

Q14

Sydney Girls

The probability density function $f(x) = \frac{1}{3}(x^2 + 1)$ is defined over the interval $[0, 1]$. Find the value of:

- $E(X)$.
- the variance.
- the standard deviation, correct to 2 decimal places.

$$\begin{aligned} \text{a) } E(X) &= \int_0^1 x f(x) dx = \int_0^1 x \left(\frac{1}{3}(x^2 + 1) \right) dx \\ &= \frac{1}{3} \int_0^1 x(x^2 + 1) dx \\ &= \frac{1}{3} \int_0^1 (x^3 + x) dx \\ &= \frac{1}{3} \left[\frac{x^4}{4} + \frac{x^2}{2} \right]_0^1 \\ &= \frac{1}{3} \left[\frac{1}{4} + \frac{1}{2} \right] = \frac{1}{4} \end{aligned}$$

$$\begin{aligned} \text{b) } \text{Var}(X) &= \int_0^1 x^2 f(x) dx - (E(X))^2 \\ &= \int_0^1 x^2 \left(\frac{1}{3}(x^2 + 1) \right) dx - \left(\frac{1}{4} \right)^2 \\ &= \frac{1}{3} \int_0^1 (x^4 + x^2) dx - \frac{1}{16} \\ &= \frac{1}{3} \left[\frac{x^5}{5} + \frac{x^3}{3} \right]_0^1 - \frac{1}{16} \\ &= \frac{1}{3} \left[\frac{1}{5} + \frac{1}{3} \right] - \frac{1}{16} = \frac{83}{720} \end{aligned}$$

$$\text{c) } \sigma = \sqrt{\frac{83}{720}} \approx 0.34 \text{ (2 d.p.)}$$

Q15

Total Education

Let X be a continuous random variable whose probability density function is $f(x) = 3x^2$ for $0 < x < 1$. Find the mean and median of X .

$$\text{mean} = \int_0^1 x f(x) dx = \int_0^1 3x^3 dx = \left[\frac{3x^4}{4} \right]_0^1 = \frac{3}{4}$$

$$\text{Let median} = a \Rightarrow \int_0^a f(x) dx = \frac{1}{2} \Rightarrow \left[x^3 \right]_0^a = \frac{1}{2} \rightarrow a = \sqrt[3]{\frac{1}{2}}$$



Continuous Probability Distribution

Part C: The normal distribution

2 unit



Q 1

Ace

The continuous random variable X has a normal distribution with a mean 11 and standard deviation 2. If the random variable Z has the standard normal distribution, then the probability that X is greater than 16 is equal to:

- (A) $P(Z > 2)$ (C) $P(Z < 2.5)$ $z = \frac{x - \mu}{\sigma} = \frac{16 - 11}{2} = 2.5$
 (B) $P(Z < -2.5)$ (D) $P(Z > 5)$

$$P(X > 16) = P(Z > 2.5) \\ = P(Z < -2.5)$$

Q 2

Ace

A factory produces bags of cashews. The weights of the bags are normally distributed, with a mean of 900 g and a standard deviation of 50 g. What is the best approximation for the percentage of bags that weigh more than 1000 g?

- (A) 0% (C) 5%
 (B) 2.5% (D) 16%

$$z = \frac{x - \bar{x}}{s} \\ = \frac{1000 - 900}{50} \\ = 2$$

95% of scores have a z-score between -2 and 2
 $\therefore$ 2.5% have a z-score greater than 2.

Q 3

Ace

Tran's industrial unit produces aluminium rods. In the past week the industrial unit has produced aluminium rods with a mean weight of 12.5 kilograms and a standard deviation of 0.5 kilograms.

- (a) Quality control requires any aluminium rod with a z-score less than -1 to be rejected. What is the minimum weight that will be accepted?
 (b) Aluminium rods with a z-score greater than 2 are also rejected. What is the maximum weight that will be accepted?

(a) $z = \frac{x - \bar{x}}{s}$
 $-1 = \frac{x - 12.5}{0.5}$
 $x = (-1 \times 0.5) + 12.5 = 12$
 $\therefore$ Minimum weight to be accepted is 12 kg.

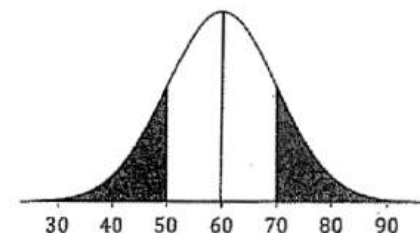
(b) $z = \frac{x - \bar{x}}{s}$
 $2 = \frac{x - 12.5}{0.5}$
 $x = (2 \times 0.5) + 12.5 = 13.5$
 $\therefore$ Maximum weight to be accepted is 13.5 kg.

Q 4

Ace

The normal distribution shows the results of a mathematics assessment task. It has a mean of 60 and a standard deviation of 10.

- (a) What is the mathematics assessment result with a z-score of -2?
 (b) What is the z-score of a mathematics assessment result of 65?



- (a) Students with a z-score of -2 is two standard deviations below the mean $(60 - (2 \times 10) = 40)$.
 $\therefore$ A score of 40 has a z-score of -2.

(b) z-score for 65
 $z = \frac{x - \bar{x}}{s}$
 $= \frac{65 - 60}{10}$
 $= 0.5$
 $\therefore$ z-score is 0.5

Q 5

Ace

The heights of a group of friends are normally distributed with a mean of 167 cm and a standard deviation of 12 cm. What percentage of the group are more than 179 cm tall?

$$z = \frac{x - \bar{x}}{s} = \frac{179 - 167}{12} = 1$$

68% of scores have a z-score between -1 and 1.
 $\therefore 32\% \div 2 = 16\%$ have a z-score greater than 1.



Continuous Probability Distribution

Part C: The Normal Distribution

2 unit

selective

Q 6

Baulkham Hills

Results for a test are given as z-scores. In this test Angela gained a z-score of 3. The test has a mean of 55 and standard deviation of 6. What was Angela's actual mark in this test?

- (A) 58
(B) 73

- (C) 64
(D) 67

$$\mu = \bar{x} = 55 \quad z = 3 \quad \therefore \text{score} = 55 + 3 \times 6 = 73$$

Q 7

Baulkham Hills

The time, T , in seconds that divers can hold their breath is normally distributed with $\mu = 120$ and $\text{Var}(T) = 400$. In what range of time length would you expect to find the middle 95%?

- (A) $100 \leq x \leq 140$
(B) $80 \leq x \leq 160$

- (C) $60 \leq x \leq 180$
(D) $40 \leq x \leq 200$

(7) $\mu = 120 \quad \sigma = 20 \quad (\text{Var}(T) = \sigma^2)$
middle 95% is 2σ around μ .

$$80 \leq x \leq 160$$

Q 8

Birrong Girls

The students in a school were surveyed on the number of hours of sleep per week. The results were normally distributed. The survey indicated that 95% of students had between 42 and 54 hours of sleep per week.

- (i) Determine the mean number of hours of sleep per week. (1 mark)
(ii) What was the standard deviation? (1 mark)
(iii) What percentage of students would have indicated they had between 51 and 57 hours of sleep per week? (2 marks)

(a) $\bar{x} = 48$

$$54 - 42 = 12 \div 4 = 3$$

$$\therefore \bar{x} = 54 - 2(3) = 48$$

(b) $\sigma = 3$

(c) $\bar{x} = 48$

% of students between 51 & 57 = 15.85%

Q 9

Blacktown Boys

The time taken to travel between two regional cities is approximately normally distributed with a mean of 85 minutes and a standard deviation of 4 minutes.

The percentage of travel times that are between 81 minutes and 93 minutes is closest to

- A. 68%
B. 70.5%
C. 81.5%
D. 95%

81 is 1 sd below a mean of 85, and 93 is 2 sd above a mean of 85.

$$68\% + \frac{95\% - 68\%}{2} = 81.5\%$$

Q10

Blacktown Boys

The chocolate consumption per person per day of a population of people was found to be normally distributed with a mean of 68.95 grams and a standard deviation of 18.45.

- a) Above what chocolate consumption rate does 2.5% of this population lie? 1
b) Rahul consumed 50.5 grams of chocolate in one day. What percentage of this population have a chocolate consumption rate more than Rahul's? 2
c) If Ben takes a sample of people from this population and finds that 6 of them consumed less than 13.6 grams of chocolate per person per day. If Ben's sample has the same distribution as this population, what is Ben's sample size? 2

a 2.5% is a 2 z-score above the mean
 $68.95 + 2 \times 18.45 = 105.85$ grams

2.5% of this population consumed more than 105.85 grams of chocolate per person per day.

b $z = \frac{50.5 - 68.95}{18.45} = -1$

$$\frac{1}{2}(100\% - 68\%) = 16\%$$

$100\% - 16\% = 84\%$
 $\therefore 84\%$ of people consumed more than 50.5 grams of chocolate per person per day.

c $z = \frac{13.6 - 68.95}{18.45} = -3$

$$\frac{1}{2}(100\% - 99.7\%) = 0.15\%$$

0.15% represents 6 people

$$\frac{6}{0.15\%} = 4000$$

$\therefore$ Sample size is 4000 people.



Continuous Probability Distribution

Part C: The normal distribution

2 unit

selective

Q11

Caringbah

If the z scores on an examination are normally distributed and $P(z \leq N) = 0.6$ for some number N , what is the value of $P(-N \leq z \leq N)$?

- A) 0.1
C) 0.2
B) 0.3
D) 0.4

$$\begin{aligned} P(-N \leq z \leq N) &= 2P(0 \leq z \leq N) \\ &= 2\{P(z \leq N) - 0.5\} \\ &= 2(0.6 - 0.5) \\ &= 2(0.1) \\ &= 0.2 \end{aligned}$$

Q12

Caringbah

The completion times for the Oztown triathlon race were normally distributed with mean times 60 minutes and standard deviation 5 minutes. Using the empirical rule, find Ozzie's completion time if he finished ahead of 84% of competitors.

2

$$\begin{aligned} 50\% + \frac{1}{2} \times 68\% &= 84\% \text{ (normal distribution)} \\ 84\% \text{ of the times are } > \mu - \sigma \\ 60 - 5 &= 55 \text{ minutes} \end{aligned}$$

Q13

CSSA

For a normally distributed set of scores, which of the following statements is FALSE?

- (A) A z -score more than 2 has a higher probability than a z -score more than 1.
(B) The frequency distribution is symmetrical about the mean.
(C) The z -score of a score describes how far that score is from the mean.
(D) The mean and the median are equal.

Q14

CSSA

A set of scores is normally distributed. A score of 12 has z -score of -2 while a score of 21 has a z -score of $+1$. Calculate the mean of the scores.

2

$$\begin{aligned} -2 \text{ to } 1 &= 3 \text{ standard deviations} \\ 3 \text{ standard deviations} &= 21 - 12 = 9 \\ \text{So each standard deviation is } 9 \div 3 &= 3 \quad (\sigma = 3) \end{aligned}$$

$$\begin{aligned} -2 &= \frac{12 - \mu}{3} \\ -6 &= 12 - \mu \\ \mu &= 12 + 6 \\ \mu &= 18 \end{aligned}$$



Q15

CSSA

Chickens are sized according to their weight. For example, a size 16 chicken weighs 1.6 kg. At a particular farm, the chickens they sell as size 15 have weights which are normally distributed with a mean of 1.56 kg and a standard deviation of 0.025 kg. Chickens that weigh less than 1.5 kg should be classified size 14 and chickens that weigh more than 1.6 kg should be classified size 16.

The following table gives the probability of scores with a z -score less than the given value.

z	first decimal place									
	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0.	0.500	0.540	0.579	0.618	0.655	0.692	0.726	0.758	0.788	0.816
1.	0.841	0.864	0.885	0.903	0.919	0.933	0.945	0.955	0.964	0.971
2.	0.977	0.982	0.986	0.989	0.992	0.994	0.995	0.997	0.997	0.998
3.	0.999	0.999	0.999	1.000	1.000	1.000	1.000	1.000	1.000	1.000

For example, the probability of a score with a z -score of less than 2.6 is 0.995, as highlighted above.

What is the probability that a randomly selected size 15 chicken has been incorrectly classified? Give your answer correct to the nearest 3 decimal places.

$$\text{Chickens } > 1.6\text{kg: } z = \frac{1.6 - 1.56}{0.025} = 1.6$$

$$\text{Chickens } < 1.5\text{kg: } z = \frac{1.5 - 1.56}{0.025} = -2.4$$

$$\begin{aligned} \text{So, the probability of incorrectly classifying a size 15 chicken is:} \\ 1 - P(-2.4 < z < 1.6) &= P(z < -2.4) + P(z > 1.6) \\ &= 0.0082 + 0.0548 \\ &= 0.063 \end{aligned}$$



Continuous Probability Distribution

Part C: The normal distribution

2 unit

selective

Q16

East Hills Boys

The students in a school were surveyed on the number of hours of sleep per week. The results were normally distributed. The survey indicated that 95% of students had between 42 and 54 hours of sleep per week.

- Determine the mean number of hours of sleep per week.
- What was the standard deviation?
- What percentage of students would have indicated they had between 51 and 57 hours of sleep per week?

a) 95% of the data lie within two standard deviations of the mean.
42 hours is a z-score of -2 and 54 hours has a z-score of 2.
The mean is midway between 42 and 54 hours.
 $\therefore$ Mean is 48 hours.

b) There are 4 standard deviations between 42 and 54 hours.
Standard deviation = $\frac{54 - 42}{4} = 3$ hours

c) To find the z-score of 51 and 57
 $z = \frac{x - \bar{x}}{s} = \frac{51 - 48}{3} = 1$ Percentage = $\frac{99.7\%}{2} - \frac{68\%}{2}$
 $z = \frac{x - \bar{x}}{s} = \frac{57 - 48}{3} = 3$ = 15.85%
 $\therefore$ 15.85% have sleep between 51 and 57 hours of sleep per week

Q17

Girraween

If the z scores on an examination are normally distributed and $P(z < N) = 0.6$ for some number N , what is the value of $P(-N < z < N)$?

- 0.1
- 0.2
- 0.3
- 0.4

Q18

Gosford

The length of steel rods produced by a machine is normally distributed with a standard deviation of 3 mm. It is found that 2.5% of all rods are less than 25 mm long. Find the mean length of rods produced by the machine.

$$\sigma = 3 \quad x = 25 \quad z = -2$$

$$\therefore -2 = \frac{25 - \mu}{3}$$

$$25 - \mu = -6 \quad \mu = 31$$

Q19

Hunters Hill

The length of a type of ant is approximately normally distributed with a mean of 5.1 mm and a standard deviation of 1.2 mm. A standardised ant length of $z = -0.5$ corresponds to an actual ant length of

- 13 mm
- 1 mm
- 4.5 mm
- 2.55 mm

$$z = \frac{x - \bar{x}}{s}$$

$$-0.5 = \frac{x - 5.1}{1.2}$$

$$x = 4.5$$

Q20

Hunters Hill

The table below shows Bob's scores in PDHPE and Biology examinations, as well as the mean and the standard deviation for each subject.

	PDHPE	Biology
μ	65	80
σ	4	3
Score	74	84

Explain which is his strongest subject, show mathematical working.

need to standardise results for comparison

$$\text{PDHPE: } z_P = \frac{74 - 65}{4} = 2.25$$

$$\text{Biology: } z_B = \frac{84 - 80}{3} = 1.33$$

greater the z score, the better his mark is. Therefore, he scored more strongly in PDHPE.





Continuous Probability Distribution

Part C: The normal distribution

2 unit



Q21

Hunters Hill

The weight, in grams, of beans in a tin is normally distributed with mean μ and standard deviation 7.8. It is known that 10.03% of tins contain less than 200 g.

- Use the Probability Table to find the value of mean value, μ . Correct your answer to 2 decimal places.
- The machine settings are adjusted so that the weight, in grams, of beans in a tin is normally distributed with mean 205 and standard deviation σ .
 - Given that 98% of tins contain between 200 g and 210 g, draw a normal distribution graph to illustrate this information.
 - Use the Probability Table to find the value of the standard deviation, σ that can be achieved with the new setting. Correct your answer to 2 decimal places.

a) From the table.

$$P(X \leq 200) \approx 0.1003 \Rightarrow z = -1.28$$

$$z = \frac{x - \mu}{\sigma}$$

$$-1.28 = \frac{200 - \mu}{7.8}$$

$$\mu = 209.984 \approx 210$$

$$ii) z_u = \frac{210 - 205}{\sigma}$$

$$z_L = \frac{200 - 205}{\sigma}$$

$$P(X \leq z_u) = 99.90 \Rightarrow z_u = 2.33$$

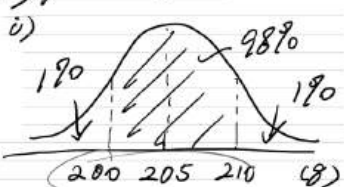
$$\frac{210 - 205}{\sigma} = 2.33$$

$$\sigma = \frac{5}{2.33}$$

$$\sigma = 2.145$$

$$\sigma = 2.15 \text{ (g)}$$

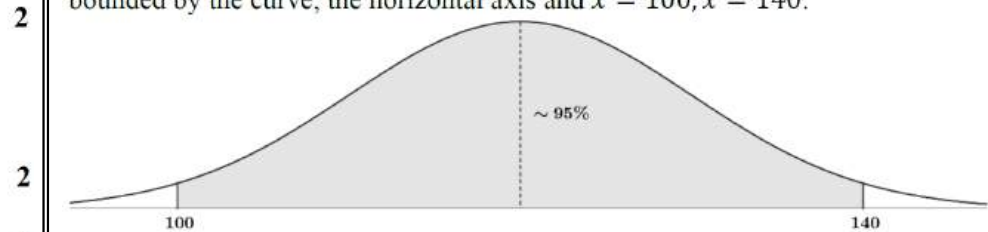
b) $\mu = 205, \sigma$



Q22

James Ruse

Consider the following graph of a normal distribution, with approximately 95% of the area bounded by the curve, the horizontal axis and $x = 100, x = 140$.



Which of the following parameters best describes the curve?

A. $\mu = 100, \sigma^2 = 140$

B. $\mu = 120, \sigma^2 = 20$

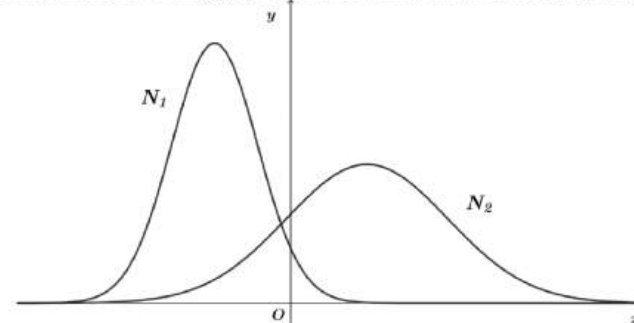
C. $\mu = 120, \sigma^2 = 10$

D. $\mu = 120, \sigma^2 = 100$

Q23

James Ruse

Consider the following graphs of two normal distributions, N_1 and N_2 .



Assuming μ_1, σ_1 are the mean and standard deviation of N_1 , and μ_2, σ_2 are the mean and standard deviation of N_2 , which of the following statements is true?

A. $\mu_1 > \mu_2$ and $\sigma_1 > \sigma_2$

B. $\mu_1 > \mu_2$ and $\sigma_1 < \sigma_2$

C. $\mu_1 < \mu_2$ and $\sigma_1 > \sigma_2$

D. $\mu_1 < \mu_2$ and $\sigma_1 < \sigma_2$



Continuous Probability Distribution

Part C: The Normal Distribution

Unit

selective

Q24

James Ruse

A probability density function f is given by

$$f(x) = \begin{cases} \frac{1}{12}(8x - x^3) & 0 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

The median m of this function satisfies which of the following equations?

- A. $m^4 - 16m^2 = 0$
 B. $m^4 - 16m^2 + 24 = 0.5$
 C. $m^4 - 16m^2 + 24 = 0$
 D. $16m^2 - m^4 - 6 = 0$

Q25

James Ruse

A pet shop has a tank of goldfish for sale. All fish in the tank may be taken to have their weights normally distributed with mean 100 g and standard deviation 10 g. Melanie is buying a goldfish and is invited to catch the one she wants. Sadly, the fish are too fast for Melanie to catch any particular fish and the one she eventually catches is done so at random.

Find the probability that the weight of the fish is:

- a) over 120 g;
 b) between 90 g and 120 g.

a) over 120 g

$$\text{Determine } z = \frac{120 - 100}{10} = 2$$

$$P(X > 120) = P(Z > 2)$$

$$= 1 - 0.9772$$

$$= 0.0228$$

$$\text{b. When } x = 90, z = \frac{90 - 100}{10} = -1$$

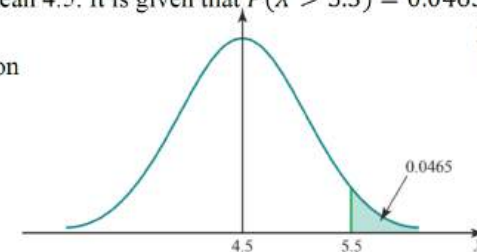
$$P(-1 \leq z \leq 2) = 34\% + 13.5\% + 13.5\% = 61\%$$

Q26

James Ruse

A random variable X is normally distributed with mean 4.5. It is given that $P(X > 5.5) = 0.0465$.

- a) Show that the standard deviation is 0.595.
 b) Find the probability that a random observation of X lies between 3.8 and 4.8.



$$\text{a) } \mu = 4.5$$

$$z = \frac{5.5 - \mu}{\sigma} \quad \text{Given } P(X > 5.5) = 0.0465$$

$$\frac{X - \mu}{\sigma} > \frac{5.5 - \mu}{\sigma}$$

$$P(Z > \frac{5.5 - \mu}{\sigma}) = 0.0465$$

$$1 - 0.0465 = 0.9535$$

z for this 1.68 from the table

$$\frac{5.5 - \mu}{\sigma} = 1.68$$

$$\sigma = \frac{5.5 - 4.5}{1.68} = 0.5952$$

$$\text{b) } P(3.8 < X < 4.8) = P\left(\frac{3.8 - 4.5}{0.595} < Z < \frac{4.8 - 4.5}{0.595}\right)$$

$$= P(-1.180 < Z < 0.500)$$

$$= 0.584$$

$$= P(0.50) + P(1.18) - 1$$

$$= 0.6915 + 0.8810 - 1$$

$$= 0.5725$$



Continuous Probability Distribution

Part C: The normal distribution

2 unit

selective

Q27

Killara

The Year 12 students at Koolara High School were surveyed on the number of hours of sleep they get per week. The results are normally distributed. The survey indicated that 95% of students had between 42 and 54 hours of sleep per week.

- Determine the mean number of hours of sleep per week.
- What percentage of students would have indicated they had between 51 and 57 hours of sleep per week?
- The Year 12 students at a nearby school, Chetswood High, were also surveyed. The results are also normally distributed with a mean of 50 hours and standard deviation of 2 hours.

Chetswood High boast they have a greater percentage of Year 12 students that get more than 54 hours of sleep per week. Are they correct? Justify your answer with relevant calculations.

$$(a) \frac{54 - 42}{4} = 3 \quad \sigma = 3 \quad \mu = 48 \text{ hours}$$

$$(b) P(x \geq 51) = \frac{(100\% - 68\%)}{2} = 16\%$$

$$P(x \leq 57) = \frac{100\% - 99.7\%}{2} = 0.15\%$$

$$P(51 \leq x \leq 57) = 16\% - 0.15\% = 15.85\%$$

Q28

Manly

The completion times for the Oztown triathlon race were normally distributed with mean time 60 minutes and standard deviation 5 minutes. Using the empirical rule, find Ozzie's completion time if he finished ahead of 84% of competitors.

Therefore must be greater than one standard deviation less than mean to be finish ahead of 84%

Therefore less than 55 minutes.

Q29

Moorebank

The weekly pay for workers at the Prosper Factory is normally distributed, with a mean of \$750 and a standard deviation of \$35.

What percentage of workers earn below \$680 a week?

- 0.15%
 - 2.5%
 - 5%
 - 47.5%

2 < \$680 means a z-score lower than -2

Therefore $100 - 50 - 47.5 = 2.5\%$ so 2.5% of people will earn below \$680.

Q30

Moorebank

Fred sits his Trial exams in Modern History and Ancient History.

The marks for the Modern History class have a mean of 54 and a standard deviation of 5.6.

The marks for the Ancient History class have a mean of 76 and a standard deviation of 2.1.

- Compare and contrast the distribution of marks for the two classes. 2
- Fred scored 65 for Modern History and 80 for Ancient History. 3
Using z-score calculations, explain which subject he performed better in and why.

- The Ancient History class has a higher mean and a smaller standard deviation, meaning the scores are grouped more closely around the mean which indicates more consistent marks. While the Modern History has a lower mean and a larger standard deviation, meaning the scores are more spread out around the lower mean, so are much less consistent marks.

$$(b) z = \frac{x - \mu}{\sigma}$$

$$\text{MH } x = 65 \quad \sigma = 5.6 \quad \mu = 54 \quad z_{\text{MH}} = \frac{65 - 54}{5.6} = 1.96428571$$

$$\text{AH } x = 80 \quad \sigma = 2.1 \quad \mu = 76 \quad z_{\text{AH}} = \frac{80 - 76}{2.1} = 1.9047619$$

Fred performed better in MH as he had a higher positive z-score meaning he performed further above the mean, compared to AH, even though he had a lower raw mark for MH.



Continuous Probability Distribution

Part C: The Normal Distribution

2 unit

selective

Q31

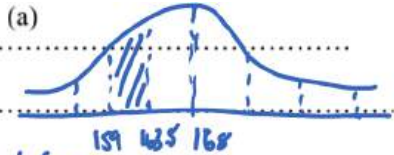
North Sydney Girls

The heights of female students at NSGHS are normally distributed with a mean of 168 cm and a standard deviation of 4.5 cm.

- (a) Find the probability that the height of a randomly selected female student attending NSGHS is between 159 cm and 163.5 cm.

It is also known that the heights of male students at NSBHS are normally distributed with a mean of 170 cm and standard deviation of 6.5 cm.

- (b) Wendy is a student at NSGHS and she is 173.5 cm tall. Caleb is a student at NSBHS and he is 176 cm tall. Who is the taller student relative to their peers in their own school? Justify your answer.
- (c) What height would Caleb need to be in order to be comparable to Wendy's height in relation to her school peers?



$$\frac{1}{2}(95\% - 68\%) = 13.5\%$$

(b) Wendy: $z = \frac{173.5 - 168}{4.5} = \frac{5.5}{4.5} = 1.22...$

Caleb: $z = \frac{176 - 170}{6.5} = \frac{6}{6.5} = 0.92...$

Since Wendy has the higher z-score, she is more standard deviations above the mean compared to Caleb so she is the taller student relative to her peers.

(c) Caleb would need a z-score of $\frac{11}{9}$

$$170 + \frac{11}{9} \times 6.5 = \frac{3203}{18} = 177.94 \approx 178 \text{ cm}$$

Q32

Sydney Girls

Use the standard normal distribution table, or otherwise, to answer the questions below.

- a) Find $P(-1.93 \leq z \leq 0.79)$. 2
- b) A large class takes an exam, and the students' scores are normally distributed. The mean score is $\mu = 72$ and the standard deviation is $\sigma = 4.5$. Students who score in the top 10% receive an A grade. What is the minimum score needed to get an A grade? 2

a)

$$P(-1.93 \leq z \leq 0.79) = P(z \leq 0.79) - P(z \leq -1.93)$$

$$= P(z \leq 0.79) - P(z \geq 1.93) = P(z \leq 0.79) - (1 - P(z \leq 1.93))$$

$$= 0.78524 - (1 - 0.97320) = 0.75844$$

b) An A-grade is greater than 90% of scores

From the table, $P(z \leq 1.28) \div 0.9$

$\therefore$ Minimum score to receive an A-grade has a z-score of 1.28:

$$z = \frac{x - \mu}{\sigma}$$

$$1.28 = \frac{x - 72}{4.5}$$

$$x = 72 + 4.5 \times 1.28$$

$$= 77.76$$



Continuous Probability Distribution

Part C: The normal distribution

2 unit



selective

Q33

Total Education

The times taken for a class of Year 12 students to complete a test are normally distributed with a mean of 37.5 minutes and a standard deviation of 1.5 minutes.

- (a) What percentage of students will complete the test in less than 34.5 minutes?
(b) 68% of students complete the test in the interval (a, b) minutes.
Find appropriate values for a and b .

2
2

(a) Let X be the time taken, and Z be the equivalent z-score

$$P(X < 34.5) = P\left(Z < \frac{34.5 - 37.5}{1.5} = -2\right) = \frac{1 - 0.95}{2} = 0.025 = 2.5\%$$

(b) Let X be the time taken, and Z be the equivalent z-score

$$P(a < X < b) = 68\% = P(-1 < Z < 1) \Rightarrow \begin{cases} a = 37.5 - 1.5 \\ b = 37.5 + 1.5 \end{cases} \Rightarrow \begin{cases} a = 36 \\ b = 39 \end{cases}$$

Q34

Western

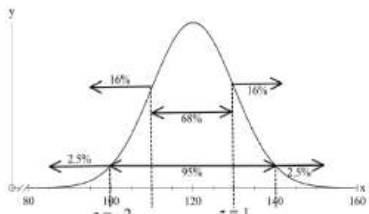
Henderson's harvests oranges to sell to Cottonworths, and the weights of the oranges they sell are normally distributed.

2

Oranges that weigh less than 100 grams are rejected, and this harvest season 97.5% of their oranges are accepted to sell.

Cottonworths also offers a bonus for premium oranges that are greater than 130 grams and 16% of this seasons harvest are classed as premium.

Find the mean and standard deviation of the weights of the oranges.



2.5 % of Oranges are rejected so since 5% lie outside the 2 sd limit
2.5% will be further than 2 standard deviations below the mean.
Therefore 100 grams lies on the -2 z-score position.

Since 16 % greater than 130 grams, similarly 16% will be at the equivalent position below (since distribution is symmetric), a total of 32 % outside these positions.
Amount between these is $100\% - 32\% = 68\%$, so these correspond to the 1 standard deviation position.
So the 16% premium Oranges are above 1 standard deviation above the mean.

So between $z = -2$ (100) and $z = 1$ (130) is 3 z-scores.

$$\text{The standard deviation must therefore be } \sigma = \frac{130 - 100}{3} = 10$$

$$\text{The mean is one z score down from } z = 1 \text{ so } \mu = 130 - 10 = 120$$

$$\therefore \mu = 120 \text{ and } \sigma = 10$$